

Randomness Pattern Hunt Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Easy	ID	randomness-pattern-hunt

Big idea: Human expectations about randomness can mistake clusters for causes and neatness for chance.

Overview

Students compare real and invented random sequences to see why randomness often looks less neat than expected.

Student Prompt

Inspect Randomness Pattern Hunt Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Randomness Pattern Hunt Stimulus A	Student-facing stimulus 1: Sequence A: H T H T T H T H H T H T; Sequence B: H H H T T H H T T T H H. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Randomness Pattern Hunt Stimulus B	Student-facing stimulus 2: Ask: which sequence looks more random, and what makes it feel that way?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Randomness Pattern Hunt Reveal / Twist	Student-facing stimulus 3: Ask: which sequence looks more random, and what makes it feel that way?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Human expectations about randomness can mistake clusters for causes and neatness for chance.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Randomness Pattern Hunt Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Randomness Pattern Hunt Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	Why do humans see meaning in random patterns?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how human expectations about randomness can mistake clusters for causes and neatness for chance.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare human-made sequences that avoid long streaks and real random sequences that include clusters.
- Ask students to commit to a judgement before revealing the source.

Ready-to-Use Examples

- Sequence A: H T H T T H T H H T H T; Sequence B: H H H T T H H T T T H H.
- Ask: which sequence looks more random, and what makes it feel that way?

Suggested Timing

Stage	Teacher move	Watch for
1	Show several short H/T or 0/1 sequences and ask students which look random.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Students mark streaks, alternations, and suspicious-looking clusters.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Reveal which sequences were random and which were invented.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Groups discuss why real randomness often includes streaks.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief the difference between pattern perception and probabilistic reasoning.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Random does not mean evenly mixed.
- Which pattern made you want to invent a cause?

Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.
- Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

- Support: use short sequences and visual color blocks.
- Challenge: introduce expected streak length in longer random sequences.

Extension Tasks

- Students generate 20 coin flips and compare them with an invented sequence.
- Connect randomness expectations to gambling, sports streaks, and data interpretation.

Related Activities

- Correlation Illusion Cards
- Proof versus Pattern
- P-hacking with Random Data